Hypothesis Testing

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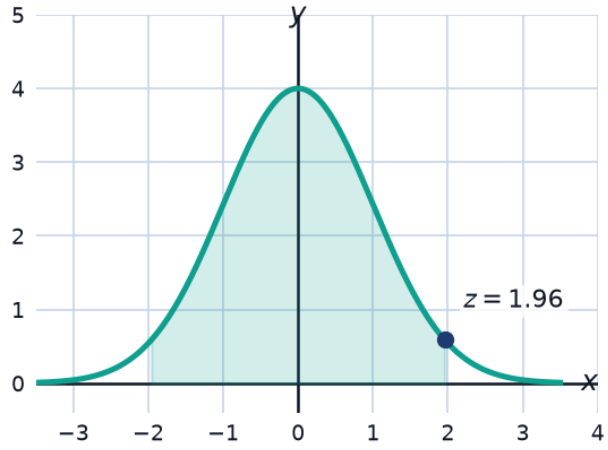
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The logic of hypothesis testing

State the null and alternative hypotheses for testing whether a new fertilizer increases the mean crop yield μ, given a historical mean $\mu_0 = 50$ kg. Classify the test as one-tailed or two-tailed.

2 [Foundational] • Target: The logic of hypothesis testing

Explain the difference between a one-tailed and a two-tailed hypothesis test, giving an example scenario for each.



3 [Foundational] • Target: The logic of hypothesis testing

A test has $H_0 : \mu = 75$ vs $H_a : \mu \neq 75$, and the sample data leads to rejecting H_0 at the 5% significance level. State, without further calculation, whether this data would also lead to rejection at the 1% level, justifying your answer.

4 [Foundational] • Target: Hypothesis test for a population mean (known σ)

A machine fills bags with a claimed mean of 500g, with known population standard deviation $\sigma = 15$g. A sample of $n = 36$ bags has mean $\bar{x} = 494$g. Test at the 5% level whether the mean is less than 500g, using $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ and critical value -1.645.

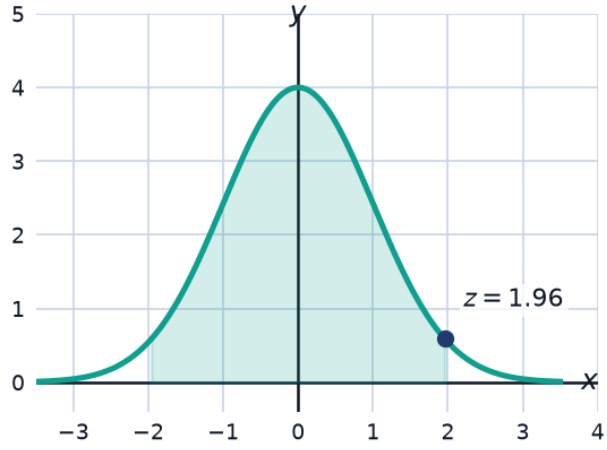
5 [Foundational] • Target: Hypothesis test for a population mean (known σ)

A population has known standard deviation $\sigma = 8$. A sample of $n = 64$ has mean $\bar{x} = 103$, against a claimed mean $\mu_0 = 100$. Test at the 1% level whether the mean exceeds 100, using critical value $z^* = 2.326$.

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6 [Foundational] • Target: Two-tailed tests for a population mean

A standardized test historically has mean $\mu_0 = 100$ with $\sigma = 15$. A sample of $n = 25$ students taking a new prep course has mean $\bar{x} = 107$. Test at the 5% level (two-tailed) whether the course changes the mean score, using critical values ± 1.96.



7 [Foundational] • Target: Two-tailed tests for a population mean

Using $z \approx 2.33$ from the previous problem, find the two-tailed p-value, given $\Phi(2.33) \approx 0.9901$, and confirm the conclusion.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Two-tailed tests for a population mean

A sample of $n = 50$ has mean $\bar{x} = 48.2$, against a claimed mean $\mu_0 = 50$ with $\sigma = 6$. Test at the 5% level (two-tailed) whether the mean differs from 50, using critical values ± 1.96.

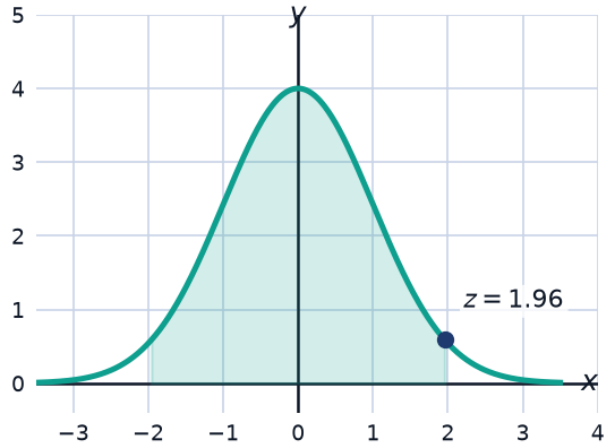
9 [Intermediate] • Target: Hypothesis test for a Poisson mean

A website historically crashes at a rate of $\lambda_0 = 4$ per month. After a software update, 1 crash is observed in a randomly selected month. Test at the 5% level whether the crash rate has decreased, using $P(X \le 1) \approx 0.0916$ for $X \sim \text{Po}(4)$.

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10 **[Intermediate]** • **Target: Hypothesis test for a Poisson mean**

Explain why an exact Poisson probability, rather than a normal approximation, is used to test a hypothesis about a Poisson mean when λ_0 is small.



11 **[Intermediate]** • **Target: Further Poisson mean tests**

A call center historically receives calls at $\lambda_0 = 6$ per hour. During a promotional event, 12 calls are received in one hour. Test at the 5% level whether the call rate has increased, using $P(X \geq 12) \approx 0.0201$ for $X \sim \text{Po}(6)$.

12 **[Intermediate]** • **Target: Further Poisson mean tests**

A factory's defect rate is historically $\lambda_0 = 3$ per shift. During a trial of new equipment, 8 defects are observed in one shift. Test at the 5% level whether the defect rate has increased, using $P(X \geq 8) \approx 0.0119$ for $X \sim \text{Po}(3)$.

13 **[Intermediate]** • **Target: Hypothesis test for a binomial proportion**

A company claims 40% of its customers use its mobile app ($p_0 = 0.4$). In a sample of $n = 15$ customers, 10 report using the app. Test at the 5% level whether the true proportion is greater than 40%, using $P(X \geq 10) \approx 0.0338$ for $X \sim B(15, 0.4)$.

14 **[Intermediate]** • **Target: Hypothesis test for a binomial proportion**

Explain why the binomial hypothesis test uses the exact binomial distribution rather than a normal approximation when n is small.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

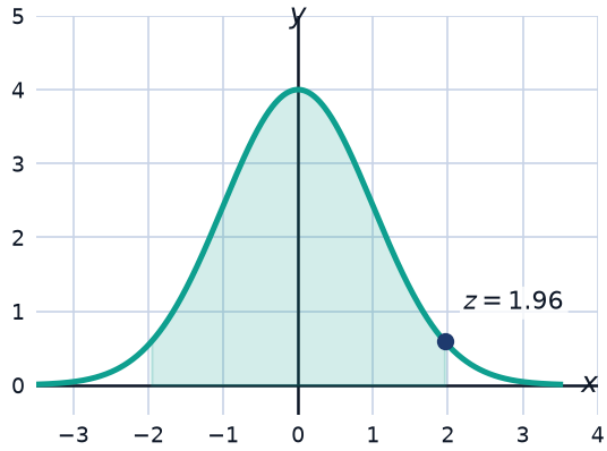
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15 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 2.01 Exponentials & Logarithms

A coin is tested for fairness ($p_0 = 0.5$). In 20 tosses, 6 heads are observed. Test at the 5% level (two-tailed) whether the coin is biased, using $P(X \le 6) \approx 0.0577$ for $X \sim B(20, 0.5)$ (doubled for a two-tailed test).

16 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 3.01 Advanced Trigonometry

A politician claims 60% of voters support a new policy ($p_0 = 0.6$). In a sample of 12 voters, only 3 support it. Test at the 5% level whether support is less than 60%, using $P(X \le 3) \approx 0.0153$ for $X \sim B(12, 0.6)$.



17 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 4.01 Differentiation

In the website crash-rate test, describe what a Type I error would mean in context, and what a Type II error would mean.

18 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 5.01 Integration

Explain what is meant by the significance level of a hypothesis test, and state the probability of a Type I error in a test conducted at the 5% significance level.

19 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 6.01 Sequences & Series

A researcher conducts a test at $\alpha = 0.10$ instead of the more common $\alpha = 0.05$. Explain how this affects the probability of Type I and Type II errors, compared to using $\alpha = 0.05$.

20 [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 7.01 Vectors

This question has four parts. A factory's current process produces components with mean breaking strength $\mu_0 = 250$ N and known $\sigma = 20$ N. A new process gives a sample of $n = 40$ components with mean $\bar{x} = 257$ N. (a) State H_0 and H_a for testing whether the new process increases mean breaking strength (one-tailed). (b) Compute the test statistic z. (c) Using critical value $z^* = 1.645$ (5% level), state the conclusion. (d) Quality control also monitors defects; the historical rate is $\lambda_0 = 5$ per batch, but a batch from the new process has only 1 defect. Using $P(X \le 1) \approx 0.0404$ for $X \sim \text{Po}(5)$, test at the 5% level whether the new process has reduced the defect rate.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Null & Alternative Hypotheses, Type I & Type II Errors

Define a Type I error in hypothesis testing.

a. Rejecting H_0 when H_0 is actually true

b. Failing to reject H_0 when H_0 is false

c. Setting the significance level too high

d. Accepting H_1 when it is false

22 [Foundational] • Target: Null & Alternative Hypotheses, Type I & Type II Errors

Define a Type II error in hypothesis testing.

a. Failing to reject H_0 when H_0 is false

b. Rejecting H_0 when H_0 is true

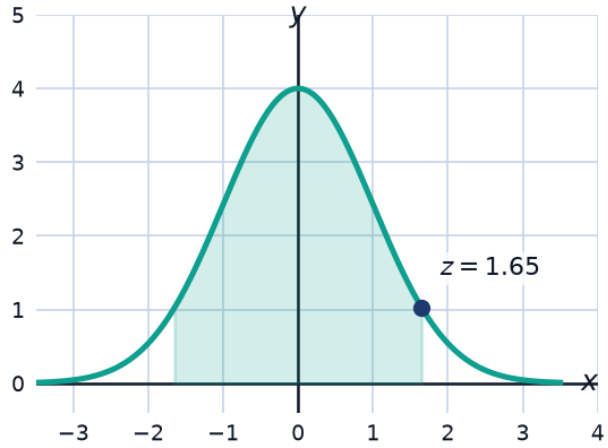
c. Computing an incorrect test statistic

d. Rejecting H_1 when H_0 is true

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23 [Foundational] • Target: Null & Alternative Hypotheses, Type I & Type II Errors

A machine produces metal pins with mean length 5.0 cm. A technician suspects the machine is out of adjustment and producing pins of different mean length. What are the appropriate hypotheses?



a. $H_0 : \mu = 5.0, H_1 : \mu \neq 5.0$

b. $H_0 : \mu = 5.0, H_1 : \mu > 5.0$

c. $H_0 : \mu \neq 5.0, H_1 : \mu = 5.0$

d. $H_0 : \mu = 5.0, H_1 : \mu < 5.0$

24 [Foundational] • Target: Testing Population Mean with Normal Distribution

A test of $H_0 : \mu = 80$ against $H_1 : \mu > 80$ at the 5% level is conducted. A sample of $n = 36$ from $N(\mu, 144)$ gives $\bar{x} = 83.5$. What is the value of the test statistic z?

a. $z = 1.75$

b. $z = 1.645$

c. $z = 2.00$

d. $z = 0.875$

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25 [Foundational] • Target: Testing Population Mean with Normal Distribution

For the test in the previous question ($z = 1.75$), what is the test conclusion at the 5% significance level (critical value 1.645)?

a. Reject H_0; there is sufficient evidence that the mean is greater than 80

b. Fail to reject H_0; insufficient evidence

c. Accept H_0; the difference is due to chance

d. Test is inconclusive

26 [Foundational] • Target: Testing Population Mean with Normal Distribution

What is the p-value for the test statistic $z = 1.75$ in a one-tailed test?

a. 0.0401

b. 0.0802

c. 0.9599

d. 0.0250

27 [Foundational] • Target: Testing Parameters with Poisson & Binomial Models

A company claims its weekly defect rate is $\lambda = 2$. In a given week, 6 defects are observed. Test $H_0 : \lambda = 2$ against $H_1 : \lambda > 2$ by calculating $P(X \geq 6)$ under H_0.

a. 0.0166

b. 0.0527

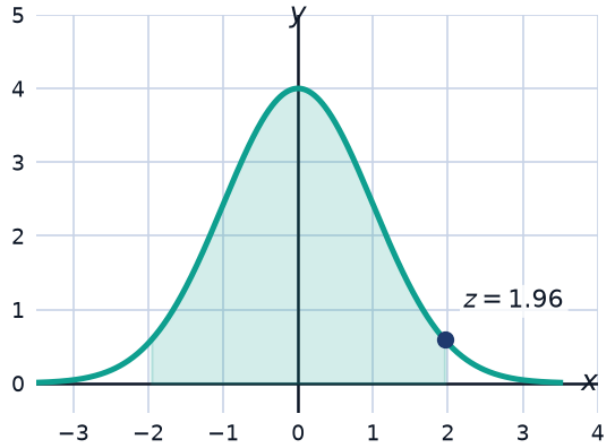
c. 0.0345

d. 0.0045

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28 [Foundational] • Target: Testing Parameters with Poisson & Binomial Models

At the 5% significance level, what is the conclusion for the defect test in the previous question ($p = 0.0166$)?



a. Reject H_0; the defect rate has significantly increased

b. Fail to reject H_0

c. Accept H_0

d. Change the significance level to 1%

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Testing Parameters with Poisson & Binomial Models

A seed packet claims a 90% germination rate. A gardener tests 20 seeds and finds 15 germinate. To test $H_0 : p = 0.9$ against $H_1 : p < 0.9$, which probability must be computed?

a. $P(X \leq 15)$ where $X \sim B(20, 0.9)$

b. $P(X = 15)$ where $X \sim B(20, 0.9)$

c. $P(X \geq 15)$ where $X \sim B(20, 0.9)$

d. $P(X \leq 15)$ where $X \sim N(18, 1.8)$

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30 [Intermediate] • Target: Critical Regions & Power of a Test

For a test of $H_0 : \mu = 50$ vs $H_1 : \mu < 50$ using $n = 25, \sigma = 10$, find the critical value for $\bar{x}$ at the 5% level ($z_c = -1.645$).

a. 46.71

b. 45.00

c. 47.50

d. 46.00

31 [Intermediate] • Target: Critical Regions & Power of a Test

In the previous test (critical region $\bar{X} < 46.71$), what is the probability of a Type II error if the true mean is $\mu = 45$?

a. 0.1964

b. 0.8036

c. 0.0500

d. 0.3085

32 [Intermediate] • Target: Critical Regions & Power of a Test

What is the statistical power of the test in the previous question (where $\beta = 0.1964$)?

a. 0.8036

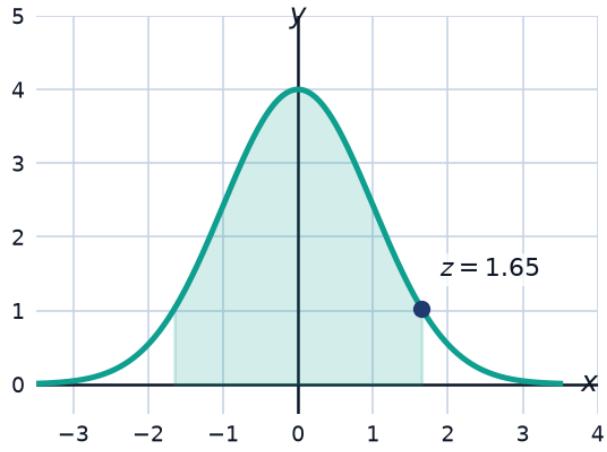
b. 0.1964

c. 0.9500

d. 0.0500

33 [Intermediate] • Target: Critical Regions & Power of a Test

Formulate the null and alternative hypotheses for the following scenario: "A factory claims the mean weight of a product is 250g. An inspector wants to test if it has changed (either direction)."



a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 3,\ H_1 : \mu > 3$

c. $H_0 : \mu = 10,\ H_1 : \mu < 10$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

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34 [Intermediate] • Target: The logic of hypothesis testing

Formulate the null and alternative hypotheses for the following scenario: "A school claims the mean test score is 70. A teacher believes it is actually higher."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$

35 [Intermediate] • Target: Hypothesis test for a population mean (known σ)

For a hypothesis test at the 1% (two-tail) significance level, find the critical z-value.

a. 1.645

b. 2.326

c. 2.576

d. 1.96

36 [Intermediate] • Target: Two-tailed tests for a population mean

For a hypothesis test at the 5% (two-tail) significance level, find the critical z-value.

a. 2.576

b. 1.645

c. 1.96

d. 1.282

37 [Intermediate] • Target: Hypothesis test for a Poisson mean

Formulate the null and alternative hypotheses for the following scenario: "A manufacturer claims the mean lifetime of a battery is 500 hours. A researcher suspects it is less than this."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 3,\ H_1 : \mu > 3$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Further Poisson mean tests • Cross-Lesson: 5.01 Integration

For a hypothesis test at the 10% (one-tail) significance level, find the critical z-value.

a. 1.282

b. 2.576

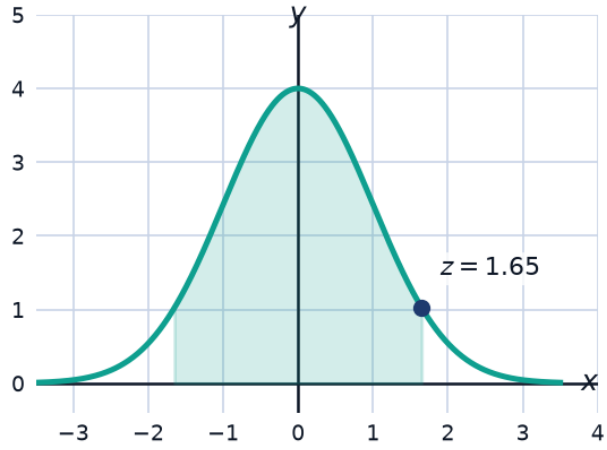
c. 2.326

d. 1.645

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39 [Challenging] • Target: Hypothesis test for a binomial proportion • Cross-Lesson: 6.01 Sequences & Series

For a hypothesis test at the 2.5% (one-tail) significance level, find the critical z-value.



a. 2.326

b. 1.96

c. 1.645

d. 1.282

40 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 7.01 Vectors

For a hypothesis test at the 5% (one-tail) significance level, find the critical z-value.

a. 1.96

b. 1.645

c. 1.282

d. 2.576

41 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A hypothesis test is carried out at the 2.5% significance level. Find the probability of a Type I error.

a. 2.5\%

b. 50\%

c. 0\%

d. 97.5\%

42 [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 9.01 Complex Numbers

For a hypothesis test at the 1% (one-tail) significance level, find the critical z-value.

a. 2.576

b. 1.282

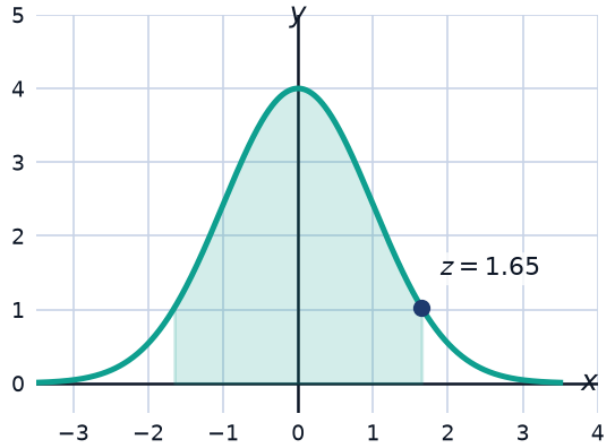
c. 1.96

d. 2.326

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43 [Challenging] • Target: Null & Alternative Hypotheses, Type I & Type II Errors • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.



a. 90\%

b. 10\%

c. 50\%

d. 0\%

44 [Challenging] • Target: Testing Population Mean with Normal Distribution • Cross-Lesson: 10.01 Numerical Methods

A hypothesis test is carried out at the 2% significance level. Find the probability of a Type I error.

a. 2\%

b. 0\%

c. 98\%

d. 50\%

45 [Challenging] • Target: Testing Parameters with Poisson & Binomial Models • Cross-Lesson: 11.01 Differential Equations

A hypothesis test is carried out at the 1% significance level. Find the probability of a Type I error.

a. 1\%

b. 50\%

c. 99\%

d. 0\%

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